20260921 Quiz

for (MATH210) Applied Complex Variables

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Exercise 18.6. Prove statement (8) in Theorem 2 of Sec. 16 using

(a) Theorem 1 in Sec. 16 and properties of limits of real-valued functions of two real variables;

(b) definition (2), Sec. 15, of limit.

Solution. Statement (8) is the sum law

$$\lim_{z\to z_0}(f(z)+F(z)) = w_0 + W_0,$$

provided that

$$\lim_{z\to z_0} f(z) = w_0, \qquad \lim_{z\to z_0} F(z) = W_0.$$

(a) Write

$$f = u + iv, \qquad F = U + iV,$$

and

$$w_0 = u_0 + iv_0, \qquad W_0 = U_0 + iV_0.$$

By Theorem 1 in Sec. 16,

$$u \to u_0, \quad v \to v_0, \quad U \to U_0, \quad V \to V_0$$

as $(x, y) \to (x_0, y_0)$. The usual sum law for real-valued functions gives

$$u + U \to u_0 + U_0, \qquad v + V \to v_0 + V_0.$$

Applying Theorem 1 again,

$$f + F = (u + U) + i(v + V) \longrightarrow (u_0 + U_0) + i(v_0 + V_0) = w_0 + W_0.$$

(b) Let $\varepsilon > 0$. There are positive numbers δ_1 and δ_2 such that

$$|f(z) - w_0| < \frac{\varepsilon}{2} \quad \text{when } 0 < |z - z_0| < \delta_1,$$

and

$$|F(z) - W_0| < \frac{\varepsilon}{2} \quad \text{when } 0 < |z - z_0| < \delta_2.$$

Set $\delta = \min\{\delta_1, \delta_2\}$. Then, whenever $0 < |z - z_0| < \delta$,

$$\begin{aligned} |(f(z) + F(z)) - (w_0 + W_0)| &\le |f(z) - w_0| + |F(z) - W_0| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

Hence the desired limit follows directly from the definition.

□

Exercise 18.8. Write $\Delta z = z - z_0$ and show that

$$\lim_{z \to z_0} f(z) = w_0 \quad \text{if and only if} \quad \lim_{\Delta z \to 0} f(z_0 + \Delta z) = w_0.$$

Solution. Since

$$\Delta z = z - z_0,$$

we have

$$z \to z_0 \quad \Longleftrightarrow \quad \Delta z \to 0,$$

and $z = z_0 + \Delta z$. Also,

$$|z - z_0| = |\Delta z|.$$

Thus the condition

$$|f(z) - w_0| < \varepsilon \quad \text{whenever} \quad 0 < |z - z_0| < \delta$$

is exactly the condition

$$|f(z_0 + \Delta z) - w_0| < \varepsilon \quad \text{whenever} \quad 0 < |\Delta z| < \delta.$$

Therefore the two limit statements are equivalent.

□

Exercise 18.9. Show that

$$\lim_{z \to z_0} f(z)g(z) = 0 \qquad \text{if} \qquad \lim_{z \to z_0} f(z) = 0$$

and if there exists a positive number M such that $|g(z)| \leq M$ for all z in some neighborhood of z_0.

Solution. Let $\varepsilon > 0$. Since $f(z) \to 0$, there is a positive number δ_1 such that

$$|f(z)| < \frac{\varepsilon}{M} \qquad \text{when } 0 < |z - z_0| < \delta_1.$$

Also, by hypothesis, there is a neighborhood of z_0 on which $|g(z)| \leq M$. Choose $\delta > 0$ small enough that both conditions hold whenever $0 < |z - z_0| < \delta$. Then

$$|f(z)g(z)| = |f(z)|\,|g(z)| \leq M|f(z)| < \varepsilon.$$

Hence

$$\boxed{\lim_{z \to z_0} f(z)g(z) = 0}.$$

□

Exercise 20.3. Using results in Sec. 20, show that

(a) a polynomial

$$P(z) = a_0 + a_1 z + a_2 z^2 + \cdots + a_n z^n \qquad (a_n \neq 0)$$

of degree n, $n \geq 1$, is differentiable everywhere, with derivative

$$P'(z) = a_1 + 2a_2 z + \cdots + n a_n z^{n-1};$$

(b) the coefficients in the polynomial $P(z)$ in part (a) can be written

$$a_0 = P(0), \qquad a_1 = \frac{P'(0)}{1!}, \qquad a_2 = \frac{P''(0)}{2!}, \qquad \ldots, \qquad a_n = \frac{P^{(n)}(0)}{n!}.$$

Solution.

(a) For every positive integer k, the monomial z^k is differentiable everywhere, with

$$\frac{\mathrm{d}}{\mathrm{d}z} z^k = k z^{k-1}.$$

The constant term a_0 has derivative zero. Thus the constant-multiple and sum rules from Sec. 20 give

$$\boxed{P'(z) = a_1 + 2a_2 z + \cdots + n a_n z^{n-1}}.$$

Thus P is differentiable everywhere.

(b) Repeated differentiation gives, for $m = 0, 1, \ldots, n$,

$$P^{(m)}(0) = m!\, a_m,$$

since every term of degree greater than m still contains a positive power of z, while terms of degree less than m have vanished. Therefore

$$\boxed{a_m = \frac{P^{(m)}(0)}{m!}} \qquad (m = 0, 1, \ldots, n),$$

with $P^{(0)} = P$.

□

Exercise 20.5. Derive expression (3), Sec. 20, for the derivative of the sum of two functions.

Solution. Suppose $f'(z)$ and $g'(z)$ exist. From the definition of derivative,

$$\begin{aligned}(f+g)'(z) &= \lim_{\Delta z \to 0} \frac{(f(z+\Delta z) + g(z+\Delta z)) - (f(z) + g(z))}{\Delta z} \\ &= \lim_{\Delta z \to 0} \left(\frac{f(z+\Delta z) - f(z)}{\Delta z} + \frac{g(z+\Delta z) - g(z)}{\Delta z} \right).\end{aligned}$$

By the sum law for limits in Sec. 16,

$$\boxed{(f+g)'(z) = f'(z) + g'(z)}.$$

□

Exercise 20.6. Derive expression (2), Sec. 20, for the derivative of z^n when n is a positive integer by using

(a) mathematical induction and expression (4), Sec. 20, for the derivative of the product of two functions;

(b) definition (3), Sec. 19, of derivative and the binomial formula (Sec. 3).

Solution. We prove

$$\frac{\mathrm{d}}{\mathrm{d}z} z^n = n z^{n-1}.$$

(a) For $n = 1$, the formula gives $(z)' = 1$. Assume that

$$(z^m)' = m z^{m-1}$$

for some positive integer m. By the product rule,

$$\begin{aligned}(z^{m+1})' &= (z^m z)' \\ &= (z^m)' z + z^m (z)' \\ &= m z^m + z^m \\ &= (m+1) z^m.\end{aligned}$$

Hence the formula follows for all positive integers n by induction.

(b) When $n = 1$, the definition immediately gives $(z)' = 1$. Now let $n \geq 2$. By the binomial formula,

$$(z + \Delta z)^n - z^n = \sum_{k=1}^{n} \binom{n}{k} z^{n-k} (\Delta z)^k.$$

Thus, for $\Delta z \neq 0$,

$$\frac{(z + \Delta z)^n - z^n}{\Delta z} = n z^{n-1} + \sum_{k=2}^{n} \binom{n}{k} z^{n-k} (\Delta z)^{k-1}.$$

Every term in the finite sum tends to zero as $\Delta z \to 0$. Therefore the definition of derivative yields

$$\boxed{(z^n)' = n z^{n-1}}$$

for all positive integers n.

□

Exercise 24.1. Use the theorem in Sec. 21 to show that $f'(z)$ does not exist at any point if

(a) $f(z) = \overline{z}$;

(b) $f(z) = z - \overline{z}$;

(c) $f(z) = 2x + ixy^2$;

(d) $f(z) = e^x e^{-iy}$.

Solution. The Cauchy–Riemann equations

$$u_x = v_y, \qquad u_y = -v_x$$

are necessary for the existence of $f'(z)$.

(a) Here $u = x$ and $v = -y$. Thus

$$u_x = 1, \qquad v_y = -1,$$

so the first Cauchy–Riemann equation fails everywhere.

(b) Since $z - \overline{z} = 2iy$, we have $u = 0$ and $v = 2y$. Hence

$$u_x = 0, \qquad v_y = 2,$$

so the first equation fails everywhere.

(c) Here $u = 2x$ and $v = xy^2$. The second equation requires

$$0 = u_y = -v_x = -y^2,$$

so $y = 0$. But then the first equation would require

$$2 = u_x = v_y = 2xy = 0,$$

a contradiction. Thus no point satisfies both equations.

(d) Since

$$f(z) = e^x(\cos y - i \sin y),$$

we have $u = e^x \cos y$ and $v = -e^x \sin y$. The Cauchy–Riemann equations become

$$e^x \cos y = -e^x \cos y, \qquad -e^x \sin y = e^x \sin y.$$

These would require $\cos y = 0$ and $\sin y = 0$ simultaneously, which is impossible.

Therefore $f'(z)$ exists nowhere in every case.

□

Exercise 24.2. Use the theorem in Sec. 23 to show that $f'(z)$ and its derivative $f''(z)$ exist everywhere, and find $f''(z)$ when

(a) $f(z) = iz + 2$;

(b) $f(z) = e^{-x}e^{-iy}$;

(c) $f(z) = z^3$;

(d) $f(z) = \cos x \cosh y - i \sin x \sinh y$.

Solution. In each case the first-order partial derivatives involved below are continuous everywhere, so the theorem in Sec. 23 applies once the Cauchy–Riemann equations are verified.

(a) Since $f(z) = (2 - y) + ix$, write $u = 2 - y$ and $v = x$. Then

$$u_x = v_y = 0, \qquad u_y = -v_x = -1.$$

Thus the Cauchy–Riemann equations hold, and

$$f'(z) = u_x + iv_x = i.$$

The derivative f' is constant, so the same theorem gives

$$\boxed{f''(z) = 0}.$$

(b) Write

$$u = e^{-x} \cos y, \qquad v = -e^{-x} \sin y.$$

Then

$$u_x = v_y = -e^{-x} \cos y, \qquad u_y = -v_x = -e^{-x} \sin y.$$

Hence

$$f'(z) = u_x + iv_x = -e^{-x} \cos y + ie^{-x} \sin y = -f(z).$$

The components of $-f$ again have continuous first partial derivatives satisfying the Cauchy–Riemann equations, so f'' exists and

$$\boxed{f''(z) = -f'(z) = f(z)}.$$

(c) Expanding

$$z^3 = (x^3 - 3xy^2) + i(3x^2y - y^3),$$

we have

$$u_x = 3x^2 - 3y^2 = v_y, \qquad u_y = -6xy = -v_x.$$

Hence

$$f'(z) = u_x + iv_x = 3(x^2 - y^2) + i6xy = 3z^2.$$

For $f'(z) = U + iV$, where $U = 3(x^2 - y^2)$ and $V = 6xy$,

$$U_x = 6x = V_y, \qquad U_y = -6y = -V_x.$$

Applying the theorem again,

$$\boxed{f''(z) = 6x + i6y = 6z}.$$

(d) Let

$$u = \cos x \cosh y, \qquad v = -\sin x \sinh y.$$

Then

$$u_x = v_y = -\sin x \cosh y, \qquad u_y = -v_x = \cos x \sinh y.$$

Hence

$$f'(z) = -\sin x \cosh y - i \cos x \sinh y.$$

For $f'(z) = U + iV$, where $U = -\sin x \cosh y$ and $V = -\cos x \sinh y$,

$$U_x = -\cos x \cosh y = V_y, \qquad U_y = -\sin x \sinh y = -V_x.$$

The first partial derivatives are continuous, so the theorem applies again and gives

$$\boxed{f''(z) = U_x + iV_x = -\cos x \cosh y + i \sin x \sinh y = -f(z)}.$$

□

Exercise 24.4. Use the theorem in Sec. 24 to show that each of these functions is differentiable in the indicated domain of definition, and also to find $f'(z)$:

(a) $f(z) = \dfrac{1}{z^4}$, $z \neq 0$;

(b) $f(z) = e^{-\theta} \cos(\ln r) + ie^{-\theta} \sin(\ln r)$, $r > 0$, $0 < \theta < 2\pi$.

Solution. We use the polar Cauchy–Riemann equations

$$ru_r = v_\theta, \qquad u_\theta = -rv_r,$$

and the derivative formula

$$f'(z) = e^{-i\theta}(u_r + iv_r).$$

All first-order partial derivatives below are continuous in the stated domains.

(a) Since

$$f(z) = r^{-4}e^{-4i\theta},$$

we have

$$u = r^{-4} \cos 4\theta, \qquad v = -r^{-4} \sin 4\theta.$$

Then

$$ru_r = -4r^{-4} \cos 4\theta = v_\theta, \qquad u_\theta = -4r^{-4} \sin 4\theta = -rv_r.$$

Thus f is differentiable for $z \neq 0$, and

$$\begin{aligned} f'(z) &= e^{-i\theta} \left(-4r^{-5} \cos 4\theta + i4r^{-5} \sin 4\theta\right) \\ &= -4r^{-5}e^{-5i\theta} = \boxed{-\frac{4}{z^5}}. \end{aligned}$$

(b) Let

$$u = e^{-\theta}\cos(\ln r), \qquad v = e^{-\theta}\sin(\ln r).$$

Then

$$ru_r = -e^{-\theta}\sin(\ln r) = v_\theta,$$

and

$$u_\theta = -e^{-\theta}\cos(\ln r) = -rv_r.$$

Hence f is differentiable throughout the stated branch domain. Moreover,

$$\begin{aligned} u_r + iv_r &= \frac{e^{-\theta}}{r}\left(-\sin(\ln r) + i\cos(\ln r)\right) \\ &= \frac{i}{r}f(z). \end{aligned}$$

Since $z = re^{i\theta}$,

$$\boxed{f'(z) = e^{-i\theta}\frac{i}{r}f(z) = i\frac{f(z)}{z}}.$$

□